

Product & Supplier Insights Report

May 6, 2026

1. Introduction

This report presents the findings from an exploratory analysis of product and supplier data from the production database. The goal was to identify data quality issues, extract meaningful insights, and provide actionable business recommendations based on the unified dataset.

2. Data Overview

The analysis was performed on a dataset comprising four key tables from the integrated catalogue:

Table	Rows
Products	10,884
SupplierProductPriceLists	14,445
SupplierProductVariants	28,897
SupplierProductStocks	22,748

3. Data Quality Assessment

During the exploration phase, several inconsistencies and quality issues were identified and addressed:

Missing Values

- Columns with more than 50% missing values (e.g., `salesUnitId`, `massUnitId`) were dropped to prevent skewed analysis.

- Remaining missing text fields were filled with "Unknown", and numerical fields were defaulted to 0 to maintain data integrity.

Duplicates

- A thorough check confirmed that there were no duplicate records across all four tables.

Outliers

- Statistical analysis flagged 280 products as high-price outliers (*above mean + 3 standard deviations*).
- Additionally, 5 products were flagged for suspiciously low prices (*below 0.10 EUR*).

4. Methodology & Feature Engineering

To prepare the dataset for deeper analysis, five meaningful features were engineered:

Feature	Description	Business Purpose
product_age_days	Days since creation	Identifies stagnant inventory vs. new arrivals
price_category	cheap / mid / expensive	Facilitates market segment analysis
in_stock	Boolean (True/False)	Quickly identifies available products for sale
supplier_prod_count	Products per supplier	Measures supplier scale and dependency
category_avg_price	Average price per category	Benchmarks pricing against category

5. Key Business Insights

Supplier Performance

- The supplier landscape is dominated by a few Large suppliers (200+ products), who contribute the bulk of the catalogue.
- Most suppliers fall into the Small segment (1–50 products).

Pricing & Market Position

- The average product price is 4.88 EUR, with the majority of the inventory categorized as "Cheap" (under 5 EUR).
- No currency inconsistencies were found, all pricing is standardized in EUR.

Inventory Status

- A majority of the products are currently in stock.
- However, the flagged outliers suggest potential data entry errors in pricing that could impact sales.

6. Strategic Recommendations

Pricing Audit

- Manually review the 280 high-price outliers and the 5 suspiciously low-priced items to ensure catalogue accuracy.

Data Collection Improvement

- Address the high percentage of missing data in attributes like brand and material to enhance the shop's filtering capabilities.

Stock Monitoring

- Implement automated alerts for "Out of Stock" items to minimize lost sales opportunities.

7. Assumptions & Limitations

- Due to the absence of direct cost data, Average Price was utilized as a proxy for margin-related analysis.
- The high-price threshold was statistically defined as mean + 3 standard deviations.
- Supplier identification was derived from the `productPropositionCategory` attribute.